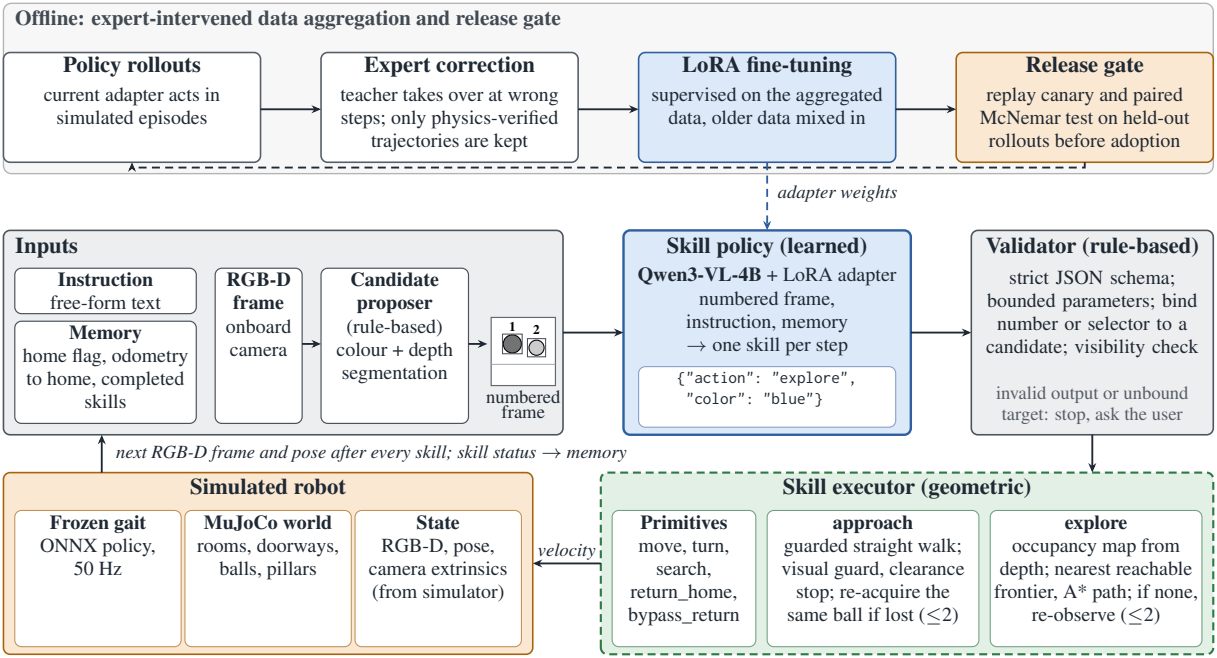

Offline: expert-intervened data aggregation and release gate
Policy rollouts
current adapter acts in simulated episodes
Expert correction
teacher takes over at wrong steps; only physics-verified trajectories are kept
LoRA fine-tuning
supervised on the aggregated data, older data mixed in
Release gate
replay canary and paired McNemar test on held-out rollouts before adoption
adapter weights
Inputs
Instruction
free-form text
Memory
home flag, odometry to home, completed skills
RGB-D frame
onboard camera
Candidate proposer
(rule-based) colour + depth segmentation
1
2
numbered frame
Skill policy (learned)
Qwen3-VL-4B + LoRA adapter
numbered frame, instruction, memory
→ one skill per step
{"action": "explore", "color": "blue"}
Validator (rule-based)
strict JSON schema; bounded parameters; bind number or selector to a candidate; visibility check
invalid output or unbound target: stop, ask the user
next RGB-D frame and pose after every skill; skill status → memory
Simulated robot
Frozen gait
ONNX policy, 50 Hz
MuJoCo world
rooms, doorways, balls, pillars
State
RGB-D, pose, camera extrinsics (from simulator)
velocity
Skill executor (geometric)
Primitives
move, turn, search, return_home, bypass_return
approach
guarded straight walk; visual guard, clearance stop; re-acquire the same ball if lost (≤2)
explore
occupancy map from depth; nearest reachable frontier, A* path; if none, re-observe (≤2)